

GPT Kya Hai?

AI Engineer Interview Guide — Basic Questions & Answers

1. GPT Kya Hai?

GPT ka matlab hai **Generative Pre-trained Transformer**. Ye ek AI language model architecture hai, aur saath hi ye OpenAI ke models (jaise GPT-3, GPT-4) ka bhi naam hai.

Simple lafzon mein, GPT ek aisa AI system hai jo insaan jaisi likhi hui zabaan samajh sakta hai aur naya text generate kar sakta hai — jaise answers dena, articles likhna, code banana, ya conversation karna.

Generative — Ye Text Banata Hai

GPT ek word ke baad dusra word predict karke text banata hai. Har naya word (ya "token") us tak ke pichle poore text ke context ko dekh kar choose kiya jata hai. Isi wajah se ye lambe, coherent aur logical jawab de sakta hai.

Pre-trained — Pehle Se Training Hoti Hai

Kisi bhi specific kaam ke liye taiyar hone se pehle, GPT ko internet ke bohat bare hisse — books, websites, articles, aur bohat sara likha hua text — par train kiya jata hai. Is training se model language ke patterns, grammar, facts, aur basic reasoning seekh leta hai.

Baad mein, is pre-trained model ko chote extra data ke sath fine-tune bhi kiya ja sakta hai taake wo specific tasks (jaise customer support ya coding) mein aur behtar ho jaye.

Transformer — Iski Building Architecture

Transformer wo neural network design hai jis par GPT bana hua hai. Ye 2017 ke ek mashhoor paper "Attention Is All You Need" mein pehli dafa introduce hua tha.

Iski sabse khaas cheez "attention mechanism" hai — ye model ko ye samajhne mein madad karta hai ke ek sentence ke andar kaunse words ek dusre se zyada related hain, chahe wo door door hi kyun na hon. Isi wajah se GPT lambay paragraphs ka context bhi acchi tarah samajh leta hai.

GPT Ka Roz Marra Use

GPT lafz do tareeqon se use hota hai. Pehla, ye ek general architecture/technique ka naam hai jo bohat si companies apne models banane ke liye istemal karti hain. Dusra, ye khaas taur par OpenAI ke models ke liye use hota hai — jaise ChatGPT, jo asal mein OpenAI ka chatbot product hai jo GPT models par bana hua hai.

Ek dilchasp baat: main khud (Claude) bhi transformer architecture par hi bana hua hoon, lekin mujhe Anthropic ne banaya hai, na ke OpenAI ne. Dono alag models hain, magar dono ki bunyaad transformer hi hai.

2. AI Engineer — Basic Interview Questions

Neeche diye gaye questions beginner se intermediate level AI/ML engineer interviews mein aksar puche jate hain. Har jawab ko simple lafzon mein samjhaya gaya hai taake asani se yaad rahe.

Q1. Machine Learning kya hai?

Machine Learning AI ki ek shaakh hai jismein computer systems data se seekhte hain aur predictions ya decisions karte hain, bina is ke ke unhein har cheez ke liye explicitly programming ki jaye. Jitna zyada data model dekhta hai, utna hi behtar patterns pehchanta hai.

Q2. Supervised aur Unsupervised Learning mein farq kya hai?

Supervised learning mein model ko labeled data diya jata hai — matlab har input ke sath uska sahi answer (output) bhi pata hota hai, jaise photos jinke sath "cat" ya "dog" ka label ho. Model isi se seekhta hai. Unsupervised learning mein data unlabeled hota hai, aur model khud data mein chupe hue patterns ya groups (clusters) dhoondta hai.

Q3. Overfitting kya hota hai aur ise kaise rokte hain?

Overfitting tab hoti hai jab koi model training data ko itni ghehrai se yaad kar leta hai ke naye, pehle na dekhe hue data par accha perform nahi kar pata — jaise koi student jo answers ratta laga le, samajhe bina. Isay rokne ke tareeqe hain: regularization (L1/L2 penalties), dropout, cross-validation, zyada training data lena, aur early stopping.

Q4. Neural Network kya hota hai?

Neural Network ek computational model hai jo insaani dimagh ke neurons se inspire hokar banaya gaya hai. Ismein layers hoti hain jo chote chote "neurons"

(nodes) par mabni hoti hain, aur ye neurons weighted connections ke zariye ek dusre se data pass karte hain taake koi pattern ya faisla seekha ja sake.

Q5. Transformer architecture kya hai?

Ye ek modern neural network design hai jo "self-attention" mechanism istemal karta hai taake input ke different hisson ki ek dusre se ahmiyat (importance) samjhi ja sake. Yehi architecture aaj kal ke tamam bare language models — GPT, BERT, Claude waghera — ki bunyaad hai.

Q6. Attention mechanism kya karta hai?

Attention mechanism model ko ye decide karne mein madad karta hai ke jab output generate ho raha ho, to input sentence ke kis hisse par sabse zyada "tawajju" (focus) deni hai. Isi ki wajah se model lambe sentences ke door door ke words ke darmiyan taluq bhi acchi tarah samajh leta hai.

Q7. Fine-tuning kya hai?

Fine-tuning ka matlab hai ke ek pehle se training hue (pre-trained) model ko kisi khaas, chote dataset par dobara thora sa train karna — taake wo kisi specific kaam mein, jaise sentiment analysis ya medical Q&A mein, aur behtar perform kare.

Q8. Bias-Variance Tradeoff kya hai?

Bias us ghalti ko kehte hain jo model ke bohat saada assumptions ki wajah se hoti hai — is se model underfitting ka shikar hota hai. Variance wo sensitivity hai jismein model data ki chote se changes se bhi bohat mutasir hota hai — is se overfitting hoti hai. Ek accha model in dono ke darmiyan balance banata hai.

Q9. Precision aur Recall mein farq batayein.

Precision batata hai ke jitne bhi predictions model ne "positive" kahi thin, un mein se kitni sach mein sahi thin (formula: $TP / (TP+FP)$). Recall batata hai ke jitne asal positive cases thay, un mein se model ne kitne sahi pehchane (formula: $TP / (TP+FN)$).

Q10. Gradient Descent kya hai?

Gradient Descent ek optimization algorithm hai jo model ke parameters ko thora thora karke adjust karta hai taake loss function (ghalti ka meter) kam se kam ho jaye. Ye har step par gradient (slope) ki opposite direction mein chalta hai, jaise pahar se neechay wadi ki taraf utarna.

Q11. Embeddings kya hote hain?

Embeddings woh tareeqa hai jis se words, sentences, ya doosri cheezein numbers ki shakal mein (vectors) represent ki jati hain. Is tareeqe mein, milte-julte maani rakhne wale words vector space mein ek dusre ke qareeb hote hain — jaise "king" aur "queen" ek dusre ke paas honge.

Q12. Tokenization kya hai?

Tokenization ka matlab hai text ko chote chote hisson (tokens) mein todna — ye tokens poore words, subwords, ya individual characters ho sakte hain. Model text ko seedha nahi samajhta, balke pehle usay tokens mein badal kar process karta hai.

Q13. RAG (Retrieval-Augmented Generation) kya hai?

RAG ek technique hai jismein AI model apna jawab generate karne se pehle kisi external knowledge base ya documents se relevant information "retrieve" (dhoond) karta hai, aur phir usay apne jawab mein shamil karta hai. Is se model ke jawab zyada sahi aur up-to-date hote hain.

Q14. Hallucination kya hota hai LLMs mein?

Hallucination us waqt hoti hai jab AI model poore confidence ke sath aisi information de deta hai jo asal mein ghalat ya khud-sakhta (made-up) hoti hai. Ye is liye hota hai kyunke model apne seekhe hue patterns ke basis par jawab banata hai, kisi database se sachai check kiye baghair.

Q15. Batch Size aur Epoch mein farq kya hai?

Batch size ye batata hai ke training ke ek step (forward aur backward pass) mein kitne data samples ek sath process kiye jaate hain. Epoch ka matlab hai ke poora training dataset ek dafa model se guzar chuka ho — yani training ka ek pura round complete ho jaye.

Tip: Interview mein har concept ko apne alfaz mein, ek chota real-world example dekar samjhane ki practice karein. Isse interviewer ko lagta hai ke aap ne cheez ko ratta nahi lagaya, balke sach mein samjha hai.